

Coding & Solution

Date	4 July 2026
Team ID	xxxxxx
Project Name	Flood Prediction System
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	FLOOD-PREDICTION-SYSTEM
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	• Data preprocessing and feature scaling using StandardScaler. • Training and evaluation of Decision Tree, Random Forest, KNN, and XGBoost models. • Selection of the best-performing model (XGBoost). • Model and scaler saved using Joblib. • Flask-based web application for real-time flood prediction. • Responsive web interface for entering weather parameters and displaying prediction results.
Pending / Incomplete Features	• Real-time weather API integration. • User authentication and role management. • Cloud deployment for public access. • Email/SMS flood alert notifications.
Setup / Run Instructions	1. Install the required libraries using pip install -r requirements.txt. 2. Open the project folder in Visual Studio Code or Jupyter Notebook. 3. Run python app.py to start the Flask server. 4. Open http://127.0.0.1:5000/ in a web browser. 5. Enter the required weather parameters and click Predict Flood to view the prediction result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project successfully demonstrates the application of machine learning techniques for flood prediction using historical weather data. The solution integrates data preprocessing, model training, evaluation, and deployment through a Flask web application, providing a simple and effective interface for real-time flood prediction. Future enhancements include cloud deployment, real-time weather data integration, and automated alert notifications.